

Wandi-SSAL Bound Value Capability Design

The Wandi-SSAL design uses bound values for many of its capabilities. For example, by establishing a bound value of 64 characters length for log message string argument, the design checks and verifies that string arguments do not exceed 64 characters. This mitigates problems where string value is not properly NULL terminated.

Bound values ensure that the logger API functions are robust, reliable and prevent abnormal behavior from impacting the application execution work flow. For example, if any of the API fails validation it simply logs an error code and returns allowing the application to continue processing.

Detailed examples on how the Bound Argument Validation works with the are provided in the `Wandi-SSAL-examples.c` program.

Dynamically Adjustable Parameters		
Bound Definition	Bound Values	Benefits
Customize generation of output log message.	Generate both development production log messages or production log message only.	Reduce generating development log messages used during development in a production environment.
Number of format string specifiers types per log message	Maximum number of format string specifiers types allowed per message is 6.	Verify that log message arguments do not exceed an unreasonable number of arguments.
Different types of format string specifiers.	Different types of supported format string specifiers are %s, %d, %f, %c, %x, %p.	Prevent use of format spring specifiers not in the list and block malicious attacker specifier format.
Length for string arguments.	Maximum length for %s string argument is 64 characters.	Avoid abnormal behavior due to unbounded, invalid string length or improperly NULL terminated string argument.
Length for message format string specification.	Maximum length format string specification is 256 characters.	Avoid abnormal behavior due to unbounded, invalid string length or improperly NULL terminated string.

Subset of ASCII characters for format string specification and string argument.	Fixed subset of ASCII characters allowed: upper/lower case alphanumeric, blank and \\, %%, \', \" and !#\$%&'()*+,-./:;<=>?@[\\]^_`{ } characters	Avoid characters not needed for message logging, such as non-printable characters.
Number of severity id=value pairs and length of value string.	Maximum number of severity levels is 4 and length of string value is 16 characters.	Use only valid pre-defined severity levels.
Number of logical message grouping id=value pairs and length of value string.	Maximum number of logical message groups is 32 and length of string value is 16 characters.	Use only valid pre-defined logical message groups.